

Miles Valencia

Local Address: 321 Steinhaus Hall University of California Irvine, CA 92697-2525

EDUCATION

University of California, Irvine (UCI) **Planned June 2027**

Doctor of Philosophy in Biology

California State University, San Bernardino (CSUSB) **July 2022**

Master of Science in Biology

Thesis: “*Early-Exercise Effects on Mice Tendon Remodeling*”

California State University, Long Beach (CSULB) **May 2018**

Bachelor of Science in Biology Education

East Los Angeles College (ELAC) **May 2015**

Associate of Arts: Natural Sciences, and Social & Behavioral Sciences - *Honors Program Certified*

RESEARCH EXPERIENCE

Dr. Manny Azizi, UCI – [muscle physiology & biomechanics lab](#) **Aug. 2022 – Present**

Graduate Researcher

- Developed novel protocol to experimentally alter *in situ* tendon materials properties using genipin and then systematically characterize properties of in-series muscles
- Mentored undergraduates in developing independent research projects, securing funding, and troubleshooting experimental protocols
- Led a team of undergraduate researchers investigating the effects of functional control on tendon mechanical properties by comparing the tail tendons of kangaroo rats and lab rats
- Designed a custom rig equipped with an ergometer to conduct materials testing on tendon fascicles while submerged in Ringer’s solution and a high-speed video camera to capture 2 orientations simultaneously
- Guided journal discussions with undergraduate students to develop their critical thinking and figure comprehension skills

Dr. Angela Horner, CSUSB – [functional morphology & biomechanics lab](#) **July 2019 – July 2022**

Graduate Researcher

- Investigated the effects of maturation and exercise on mice tendon remodeling using a mice colony artificially selected for high-voluntary wheel running behavior called [High-Runners](#)
- Conducted a pilot study to assess the effects of different training protocols on mice tendon materials properties
- Drafted custom code in RStudio for importing, wrangling, and plotting exploratory graphs concerning mice wheel training
- Housed 2 cohorts of 60 mice consisting of normal and High-Runner linetypes, and recorded the wheel activity of 20 mice simultaneously
- Measured mice tendon morphometrics using an image processing software (ImageJ, NIH), measured forces using an ergometer and physiological data analysis software (ADInstruments LabChart), recorded high-speed videos of materials testing on tendon to visually track mechanical strain (ProCapture, XCitex), and finally, conducted data analysis using IGOR Pro
- Performed analyses using the statistical program JMP and produced data visualization using custom code in RStudio
- Collaborated with computer science researchers to design a 3D printable jump platform to quantify jump frequencies of mice in separate cages using infrared sensors and a raspberry Pi
- Wrote, revised, and completed a master’s thesis titled: “Early-Exercise Effects on Mice Tendon Remodeling”

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Dr. Sandy Kawano, CSULB – [Fins and Limbs Lab](#)

Hispanic Opportunities for Graduate Access and Retention (HOGAR) Program **Mar. 2019 – July 2019**
Post-Baccalaureate Researcher

- Improved the protocols of 2 teaching labs for Comparative Animal Physiology
- Applied an array of water temperatures to marine ectotherms to show the cardiovascular responses of animals with different physiologies
- Developed a teaching lab investigating the effects of neurotransmitters on the cardiovascular system utilizing the transparent carapace of Ghost Shrimp and affordable digital microscopes
- Conducted a literature review on the physiologies of marine ectotherms and incorporated relevant studies into the teaching lab protocols
- Collaborated with team to improve the lab activity's readability for students
- Identified the Student Learning Outcomes, and verified that the lab activity achieved them
- Coauthored a teaching article about ectotherm physiology

Volunteer Researcher

June 2018 – Mar. 2019

- Investigated the locomotor strategies of Spanish ribbed newts that primarily live in semi-aquatic conditions, so we may compare the differences in 3D kinematics with studies on terrestrial salamanders and better understand the water-to-land transition in vertebrate evolution
- Tracked 6 landmarks on the forelimb and pectoral girdle using XMA Lab and calculated joint angles, duty factor, and speed using custom code in RStudio
- Performed linear mixed model effects using RStudio to calculate statistical differences between the kinematic variables of the semi-aquatic and terrestrial salamanders

Undergraduate Research Assistant

Sep. 2017 – May 2018

- Investigated the locomotor strategies utilized by African mudskippers for different environmental features, which then informed models for the water-to-land transition in vertebrate evolution
- Tracked 3-4 landmarks on amphibious fish using the DLTdv5 GUI in MATLAB; these landmarks were converted into 3D kinematics where we determined displacement, bending of the body, and the degree of fin protraction and retraction
- Coordinated animal husbandry and assisted with live animal care for the Dr. Kawano's research and teaching labs
- Conducted mock peer reviews of manuscripts with 3 lab members

Dr. Ted Stankowich, CSULB – [Mammal Lab](#)

June 2018 – Aug. 2018

Volunteer Researcher

- Dissected cranial muscles from coyotes to measure the differences in biting forces of rural and urban coyotes by estimating the bite muscles of the jaws through anatomical dissections

PUBLICATIONS

1. **Valencia M**, Chung K^U, Plascencia B^U, Azizi M, "Genipin enhances *ex vivo* tendon mechanical properties regardless of age" (2025*). *In prep*.*.
2. **Chow H^U**, Valencia M, Yamauchi E^U, Moreno D^U, McGowan C, Azizi M, "Tale of two tails: material properties of kangaroo rat (*Dipodomys deserti*) and lab rat (*Rattus sp.* BNx344) tail tendon fascicles" (2025*). *In prep*.*.
3. **Valencia M**, Monroy J, Garland T, Horner A, "Exercise effects on tendon in mice selectively bred for high running" (2025*). *Physiological Reports. In revision**.
4. **Valencia M**, "Early-Exercise Effects on Mice Tendon Remodeling" (2022). *Electronic Theses, Projects, and Dissertations*. 1554. <https://scholarworks.lib.csusb.edu/etd/1554>

^U = Undergraduate coauthor

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TEACHING EXPERIENCE

UCI School of Biological Sciences

Pedagogical Scholars Program

Jan. 2025 – Present

UNI STU 390A & B – *Advanced Pedagogy and Academic Job Preparation* (2 quarter)

- Compared recent pedagogical research discussing strategies for implementing evidence-based practices
- Developed lesson plans for introducing and preparing graduate students to teach as new instructors

Graduate Teaching Assistant

Sep. 2022 – Present

BIO SCI 100 – *Scientific Writing* (1 quarter)

- Guided discussions with students to better comprehend research hypotheses and figures

BIO SCI D170 – *Applied Human Anatomy* (5 quarter)

- Facilitated active learning strategies like jigsaw and speed-dating group activities

BIO SCI E109 – *Human Physiology* (1 quarter)

- Assisted in processing and importing PollEverywhere surveys using custom R code

BIO SCI E112L – *Physiology Lab* (1 quarter)

- Facilitated experiments that used BIOPAC data acquisition systems to explain physiological concepts

CSUSB College of Natural Sciences

Sep. 2019 – July 2022

Graduate Teaching Assistant

BIOL 100 Lab – *Topics in Biology* (1 quarter)

- Tailored lectures about the scientific method and core biology concepts to the interests and applications for non-biology major undergraduates
- Guided classrooms of 24 students through exercises such as microscope slide preparation

BIOL 201 Lab – *Biology of Organisms* (1 quarter)

- Designed lectures about organismal diversity that promoted a class culture of student engagement
- Taught proper microscopy techniques including gram stains to classrooms of 24 students

BIOL 224 Lab – *Human Anatomy and Physiology II* (1 quarter)

- Produced and edited online lectures using Zoom and Camtasia
- Taught students about different organ systems (i.e. the peripheral circulatory system)

BIOL 2230 & 2240 Lab – *Human Anatomy and Physiology I & II* (4 semesters)

- Developed lectures incorporating physical and virtual models for teaching Anatomy and Physiology through a virtual classroom (i.e. AnatomyStandard.com)
- Replicated an in-person experience for students by integrating group work using tools such as zoom breakout rooms, BlackBoard discussion boards, and Google JamBoards.
- Demonstrated cat dissections while explaining musculoskeletal elements

CSULB College of Natural Sciences

Jan. 2018 – May 2018

Laboratory Instructor's Aide

BIOL 213 Lab – *Biology: Introduction to Ecology & Physiology* (1 semester)

- Corrected and provided feedback for quizzes
- Presented two lectures on ecology and the cardiovascular system to sophomore and juniors
- Assisted and demonstrated techniques in laboratory exercises and dissections

PROFESSIONAL EXPERIENCE

Coyote Research Ambassador

Aug. 2020 – May 2021

Office of Student Research, CSUSB

- Informed undergraduate classes about college resources that help make research accessible
- Peer mentored undergraduates to help achieve their research goals (i.e. construct a research poster)

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- Co-/Hosted virtual workshops to 60+ audience members about research topics such as getting involved with research, and grant proposal writing
- Developed a workshop for preparing graduate students to be successful in their graduate programs
- Hosted faculty interviews on Instagram to increase outreach to the student population
- Participated in a search committee for a new director for the Office of Student Research

Biology and Mathematics Tutor

Oct. 2018 – Oct. 2019

Varsity Tutors, Online Platform

- Networked with parents to gain biology and mathematics students
- Organized appointments considering tutoring locations as well as students' time restraints
- Built a strong rapport with students and parents by showing my mastery of subjects and altering my tutoring style according to the students' needs

Tutor Manager

Oct. 2016 – May 2019

Athletic Boosters, Christian Valley High School

- Tutored 9th through 12th grade student athletes, - predominantly from African American and Latino communities
- Organized the tutoring schedule for students, parents, and tutors
- Reinforced concepts taught in class from Algebra I to Trigonometry by modeling problem-solving techniques and helping students practice with examples
- Explained scientific concepts for classes such as Biology and Physiology

English and Writing Tutor

Jan. 2014 – Jan. 2017

Writing Center, ELAC

- Assessed the learning performance of students and then modified the curriculum accordingly to improve their educational experiences
- Guided students with their ideas to correctly answer their prompts and fostered their confidence to attempt independent progress with their writing
- Taught grammar and writing workshops to classes up to 15 students
- Assessed co-tutors' skills and provided constructive feedback to improve tutoring capabilities

MENTORING & OUTREACH

Research Discovery Mentor, UCI

Jan. 2025 – Present

Undergraduate Research Opportunities Program

- Mentored 4 undergraduate freshmen in pursuing research
- Demystified the research process from joining a research lab to publishing peer-reviewed articles

Near-Peer Mentor, UCI

July – Sep. 2024

Integrative Movement Sciences Institute

- Mentored 4 undergraduate researchers involved with 3 independent projects
- Trained mentees on materials testing protocols including field-specific programs and equipment

Professional Development Chair, UCI

Aug. 2023 – Present

UCI Chapter of SACNAS

- Coordinated MENTORÍA, a mentoring program designed to cultivate a sense of community among underserved students and empower students to reach their goals
- Connected with resource centers on campus to organize joint events addressing the needs underrepresented minorities; events include Getting Involved with Research, Student Organization Panel, Applying to Graduate School, Community Building in STEM, and more
- Mentored undergraduate board members in creating and coordinating professional development workshops

Graduate Panel on *Origin Stories*, CSUSB

Nov. 2023

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Best Student Poster Presentation Judge, SWOB

Nov. 2023

Graduate Research Mentor, UCI

Sep. – Dec. 2023

Transfer-to-Teaching NOYCE Scholarship Program

- Demystified research for future educators by highlighting current research in the lab
- Guiding mentees through data collection and analysis for materials testing research

Science Writing Integrated Mentoring (SWIM) Program

Fall 2022 & 2024

Ecology and Evolutionary Biology Department, UCI

Sorte et al. (2020), “A Hierarchical Mentoring Program Increases Confidence and Effectiveness in Data Analysis and Interpretation for Undergraduate Biology Students.”

<https://www.lifescied.org/doi/full/10.1187/cbe.19-10-0201>

- Guided 5 students from generating research ideas to finalizing their individual research proposals
- Revised and provided comments to improve students’ research proposal drafts

Panel Discussion on *Students Accepted into Graduate Programs*, CSUSB

Apr. 2021

- Discussed the difficulties of applying to graduate programs, and shared the lessons learned throughout my master’s program and going into my PhD program

AFSA’s Education Committee

Jan. 2021 – Jan. 2023

- Collaborated with committee in organizing and coordinating projects that increase awareness of Filipino science (i.e. Kumustahans, podcasts and Youtube series highlighting Filipino researchers)

AFSA Mentorship Program, Online

Oct. 2020 – Jan. 2023

- Shared about our Filipino identities as well as career & personal goals
- Identified application requirements for fellowships and graduate schools in the Philippines
- Coordinated 2nd-to-present iterations of program by collaborating with team about mentor-mentee pairs and organizing guest speakers for skill-building workshops

National Biomechanics Day at CSULB

Apr. 2019

- Coordinated with the movement science and robotics labs to introduce students to biomechanics
- Engaged students in a biomechanics activity on lever systems using vertebrate skeletons
- Emphasized the overlapping areas of biorobotics, human and animal biomechanics

Jordan High School visit at CSULB

Nov. 2018

- Presented ongoing research projects in Dr. Sandy Kawano’s lab to high school seniors
- Engaged students in a biomechanics activity on lever systems using vertebrate skeletons
- Informed groups about applications of biomechanics research to the medical field

GRANTS, AWARDS, AND FELLOWSHIPS

UCI Integrative Movement Sciences-Summer Research Institute (\$500)	Jun. 2025
UROP Research Discovery Program Mentor (\$1500)	Jun. 2025
UCI Ecology and Evolutionary Biology: SWIM Mentorship (\$500)	Dec. 2024
GAANN Fellowship AY 2024-2025 (\$6826)	Dec. 2024
UCI Integrative Movement Sciences-Summer Research Institute (\$500)	Jun. 2024
UCI Ecology and Evolutionary Biology Travel Award (\$500)	July 2024
GAANN Research Funds (\$3000)	June 2024
GAANN Fellowship Summer 2024 (\$8631)	June 2024
UCI Associated Graduate Students Travel Award (\$250)	Aug. 2023
UCI DTEI Graduate Scholars Program (\$5000)	June 2023
UCI Ecology and Evolutionary Biology Travel Award (\$500)	May 2023
UCI EEB Department nomination for Faculty Mentor Fellowship	Mar. 2023, 2024
UCI Ecology and Evolutionary Biology: SWIM Mentorship (\$500)	Dec. 2022
CSUSB College of Natural Sciences: Outstanding Graduate Student Award	May 2022

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CSUSB Office of Student Research: Student Travel Award (\$819.90)	Jan. 2022
SICB Broadening Participation Travel Award (\$300)	Jan. 2022
UCI Integrative Movement Sciences-Summer Research Institute (\$2000)	Jul. 2021
CSUSB Outstanding Graduate Student Researcher (\$500)	Apr. 2021
CSUSB Office of Student Research: Research Supplies Grant (\$1000)	Mar. 2020
SICB Charlotte Mangum Travel Award	Jan. 2019, 2020
SICB Broadening Participation Travel Award (\$500 x2)	Jan. 2019, 2020
Cumulative funds – \$33,826.90	

funds awarded to undergraduate students whom I provided application revisions

UCI UROP Summer Undergraduate Research Program (\$1500 x2)	May 2025
UCI UROP Travel Award (\$500 x2)	Jan. 2025
UCI UROP Research Experience Fellowship (\$400)	Dec. 2024
UCI UROP Summer Undergraduate Research Program (\$1500 x2)	June 2024
UCI UROP Travel Award (\$1000)	Dec. 2023
UCI UROP Research Experience Fellowship (\$1000)	Dec. 2023
UCI UROP Summer Undergraduate Research Program (\$1500 x2)	June 2023
CSUSB Undergraduate Summer Research Program (\$2800 x3)	June 2021
Cumulative funds – \$18,000	

PROFESSIONAL AFFILIATIONS

Society for the Advancement of Biology Education Research (SABER) <i>Attended SABER West Conference in January 2024</i>	Jan. 2024 – Present
American Physiological Society (APS) <i>Attended National American Physiology Summit Conference in April 2023</i>	Dec. 2022 – 2024
Society of Integrative and Comparative Biology (SICB) <i>Attended National SICB Conference in January 2018-2020, 2022-2025</i> <i>Attended Southwest Regional Meetings of Organismal Biologists (SWOB) in November 2018, 2023-2024</i>	Nov. 2017 – Present
Association of Filipino Scientists in America (AFSA)	Oct. 2020 – Present
Society for Advancement of Chicanos/Hispanics and Native Americans in Science (SACNAS)	Aug. 2021 – Present

INVITED TALKS AND GUEST LECTURES

Guest Lectures

UCI D170 – Applied Human Anatomy <i>Heart Structures and Blood Flow</i>	May 2023
PowerPoint lecture incorporated with think-pair-share group activities	
<i>Muscles and Joints</i>	May 2023
Short lecture followed by groupwork to understand how muscles act on joints	
CSUSB Biology 4240 – Animal Physiology <i>Tendons</i>	Nov. 2021
PowerPoint presentation explaining tendon anatomy & physiology and my master's research	

Invited Talks

CSUSB Office of Student Research Undergraduate Summer Research Program	July 2021
Opening talk sharing my academic journey with undergraduate researchers followed by a Q&A	

ABSTRACTS AND PRESENTATIONS

American Association for the Advancement of Science, Pacific Division

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- Talk: “A tale of two tails: material properties of kangaroo rat and lab rat tendons” **June 2024**
- Winter Ecology and Evolutionary Biology Graduate Student Symposium**
- Talk: “Regional variation in tendon fascicle properties.” **Mar. 2024**
- SICB National Meeting**
- Talk: “Muscle enhancing drug? Effects of genipin on muscle performance” **Jan. 2025**
- Talk: “Regional variation in tendon fascicle properties” **Jan. 2024**
- Talk: “Effects of wheel training intensity on tendon mechanics” **Jan. 2023**
- Talk (online): **Jan. 2022**
“Jumping ahead of aging: exercise effects on early life mice tendon remodeling”
- Poster: **Jan. 2020**
“The effects of different exercise regimens on tendon remodeling in mice (*Mus musculus*)”
- Poster: “Comparative kinematics of the forelimb during **Jan. 2019**
terrestrial locomotion in semi-aquatic versus terrestrial salamanders”
- SICB Regional Meeting, SWOB**
- Talk: “Effects of genipin on muscle-tendon dynamics” **Nov. 2024**
- Poster: “Biomechanical comparisons of forelimb **Nov. 2018**
function in walking semi-aquatic and terrestrial salamanders”
- APS National Meeting** **Apr. 2023**
- Poster: “Wheel running intensity effects on mice tendon mechanics.”
- Society of Experimental Biology Virtual Conference** **July 2021**
- Talk: “Quantification of Jump Activity in Rodent Home Cages Using Wireless IR Sensors”
- CSUSB Student Research Competition - 1st place in Bio. and Agricultural Sciences** **Feb. 2021**
- Talk: “Exercise training to increase stress?”
- CSUSB Meeting of the Minds, San Bernardino, California** **May 2020**
- Talk: “Exercise training to increase stress?”

PROFESSIONAL TRAINING

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- Inclusive Mentoring Certificate – Integrative Movement Sciences Institute** **Sep. 2024**
 - Learned about navigating mentoring relationships considering mental health, mentee understanding, healthy boundaries, and more.
 - Graduate Leadership – Course 203B** **Jan. – Mar. 2024**
 - Co-organized the Winter Ecology and Evolutionary Biology Graduate Student Symposium where students from the department present oral and poster presentations of their research
 - Collaborated on future departmental projects addressing department culture, program requirements, improvements to graduate courses, and more
 - Using Classroom Observation Protocol Data to Document Professional Growth in your Teaching SABER West Workshop** **Jan. 2024**
 - Learned about the Classroom Observation Protocol for Undergraduate STEM (COPUS) to characterize how faculty and students are spending their time in the classroom
 - Practiced using COPUS method to quantify and evaluate teaching practices
 - Certificate in Mentoring Excellence – UCI Graduate Division** **Nov. 2023**
 - Learned about effective mentoring practices including topics covering mentoring philosophies, identity salience, conflict resolution, and resilience.
 - Certificate in Inclusive Course Design**
 - UCI Division of Teaching Excellence and Innovation** **Aug. 2023**
 - Assisted in converting a 300+ student class from an in-person to hybrid format
 - Discussed how to incorporate specifications grading into a course design

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Certificate of Teaching Excellence

UCI Division of Teaching Excellence and Innovation

May 2023

- Applied pedagogy approaches taught in University studies 390X within a classroom setting
- Observed 3 classes and provided constructive feedback for the presenters, and vice versa 3 times

Intro to Higher Ed Research Workshop Series

UCI Division of Teaching Excellence and Innovation

Jan. – Mar. 2023

- Learned about qualitative and quantitative approaches to address pedagogical questions
- Developed a comprehensive plan to study how reducing prerequisites affect pre-nursing student success in Anatomy and Physiology courses at CSUSB

University Studies 390X: Developing Pedagogy – UCI

Aug. – Dec. 2022

- Learned about a variety of teaching topics including active learning, promoting inclusivity and accessibility, assessment types and selection, instructional technologies, and more
- Applied pedagogical approaches within simulated classes and discussed feedback from observers
- Designed a course syllabus for Anatomy and Physiology

Integrative Movement Sciences – UCI Summer Research Institute

July 2021

- Collaborated with interdisciplinary scientists tackling biomechanical questions from different organizational scales
- Addressed how optimal control modeling could predict locomotor strategies of kangaroo rats competing for food resources

Issues-X Online Teaching Workshop – CSUSB

July 2020

- Practiced applying computer applications for implementing group activities in a virtual setting (i.e. Google JamBoards)
- Learned different ways to encourage and maintain student engagement (i.e. Breakout rooms)

Broadening Participation Meeting Mentorship Program – SICB

Jan. 2018 – 2020, 2022

- Acknowledged academic weaknesses with mentor and discussed exercises to strengthen them
- Coordinated meetings with students and faculty from potential labs for graduate school
- Discussed plans for success such as applying for the NSF Graduate Research Fellowship Program